SKILL VALIDATION AND HIRING PLATFORM

A MINI PROJECT REPORT

Submitted by

ADISHESH L (MBC23CS011)

ANGEL MARY JOSE (MBC23CS026)

ARDHRA ANNA MATHEWS (MBC23CS029)

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to

APJ Abdul Kalam Technological University in fulfilment
of the requirements for the award of the Degree
of
Bachelor of Technology
in
Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY

PEERMADE-685531

APRIL,2026

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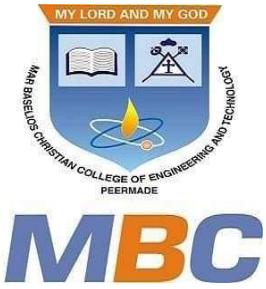
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PEERMADE

CERTIFICATE

This is to certify that report entitled "**SKILL VALIDATION AND HIRING PLATFORM"** submitted by Adishesh L (MBC23CS011), Angel Mary Jose (MBC23CS026), Ardhra Anna Mathews (MBC23CS029),**Ebin Abraham** (MBC23CS049) APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Prof. Nissu Simon
Project Guide
Assistant Professor
Department of CSE
MBCCET

Dr. Ushus Maria Joseph
Head of the Department,
Department of CSE,
MBCCET

ACKNOWLEDGEMENT

We take this opportunity to express our sincere gratitude to everyone who supported and guided us throughout the successful completion of our Mini Project– "**Skill Vaildation and Hiring Platform**".

First and foremost, we thank the **Almighty God** for His blessings and guidance, which gave us the strength and determination to complete this phase of our project successfully.

We express our heartfelt gratitude **to Dr. V. I. George**, Principal, **Mar Baselios Christian College of Engineering and Technology, Peermade**, for providing us with the facilities and encouragement to carry out our project work.

We are extremely indebted to our Head of the Department, **Dr. Ushus Maria Joseph** , Department of Computer Science and Engineering, for her valuable suggestions and encouragement during our project work.

We are grateful to thank my project guide, **Prof. Nissu Simon,** Assistant Professor, Department of Computer Science and Engineering for his valuable support.

We owe our deep gratitude to our project coordinator, **Prof. Nissu Simon,** Assistant Professor, Department of Computer Science and Engineering who took a keen interest in our project and guided us all along, till the completion of our project by providing all the necessary information for developing a good presentation. We are thankful to and fortunate enough to get constant encouragement, support, and guidance from all the faculty members of the Department of Computer Science and Engineering, which helped us in completing our project work.

We extend our gratitude to all the faculty members of the Department of Computer Science and Engineering for their cooperation and assistance.

Finally, we thank our families, classmates, and friends for their constant encouragement, valuable suggestions, and moral support, which inspired us to successfully complete this phase of our project .

ADISHESH L (MBC23CS011)

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EBIN ABRAHAM (MBC23CS049)

DECLARATION

We hereby declare that this report entitled "**SKILL VALIDATION AND HIRING PLATFORM"** is a bonafide record of the preliminary project work conducted by us under the supervision of **Prof. Nissu Simon,** Assistant Professor, in the Department of Computer Science and Engineering, Mar Baselios Christian College of Engineering and Technology, Peermade. The project proposal presented in this report has been presented for the award of any other degree.

The submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referred the original sources.

We also declare that we have adhered to academic ethics and integrity and have not misrepresented or fabricated any data, idea, or source in this report. We understand that any violation of the above will be subject to disciplinary action by the institution and/or the University.

This report has not previously been submitted for the award of any degree, diploma, or similar title in any other University or institution

Peermade

Date: 7th April 2026

ADISHESH L (MBC23CS011)

ANGEL MARY JOSE (MBC23CS026)

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INSTITUTION VISION

An Engineering Institute with global quality to groom competent engineers equipped to address the changing needs of society.

INSTITUTION MISSION

M1: Infrastructurel Relevance :

Develop, maintain and manage our campus for our stakeholders.

M2: Life Long Learning:

Encourage our stakeholders to participate in lifelong learning through industry and academic interactions.

M3: Social Connect:

Organize socially relevant outreach programs for the benefit of humanity.

DEPARTMENT VISION

To become an excellent department by empowering the graduates to address the challenges in the fast-changing scenario, adhering to ethical values.

DEPARTMENT MISSION

M1: Inculcate skills to solve complex engineering problems and instil entrepreneurial capabilities.

M2: Equip the students to advance with the recent trends and technology through Research and Innovations in the field of computer science.

M3: Facilitate the development of professional behaviour, ethical values and the spirit of service to the society.

PROGRAMME EDUCATIONAL OBJECTIVES

Our Graduates will be able to:

PEO1: Think creatively and critically to solve problems in a global context with varying

complexity.

PEO2: Exhibit leadership qualities, Ethical Values, managerial and entrepreneurship skills.

PEO3: Contribute inventions and innovative advancements in information technology through

life-long learning in a sustainable manner.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Apply theoretical and practical knowledge to analyse and model hardware systems withvarying complexity.

PSO2: Utilize modern programming platforms, tools and logical reasoning skills to analyse and develop application software.

PSO3: Apply computational and networking skills to solve and develop secure communication environment.

POs &PSOs

PROGRAM OUTCOMES (POs)

Engineering Graduates will be able to:

PO1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem Analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics,natural sciences, and engineering sciences.

PO3. Design/ Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6. The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need
for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

COURSE OUTCOMES (COs)

After successful completion of the course, the students will be able to:

CO1	Identify technically and economically feasible problems.
CO2	Identify and survey the relevant literature for getting exposed to related solutions and get familiarized with software development processes.
CO3	Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions of minimal complexity by using modern tools & advanced programming techniques.
CO4	Prepare technical report and deliver presentation.
CO5	Apply engineering and management principles to achieve the goal of the project.

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3.0	3.0	2.0	2.0	-	2.0	1.0	1.0	1.0	1.0	1.0	1.0
CO2	2.0	2.0	-	1.0	1.0	1.0	-	1.0	1.0	1.0	1.0	1.0
CO3	2.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	1.0	-	1.0	1.0
CO4	2.0	2.0	2.0	3.0	1.0	-	-	1.0	1.0	1.0	1.0	1.0
CO5	2.0	2.0	2.0	2.0	1.0	2.0	2.0	-	1.0	-	1.0	1.0
CO6	2.0	2.0	2.0	1.2	1.0	1.0	1.5	1.0	1.0	1.0	1.0	1.0

SUSTAINABLE DEVELOPMENT GOALS (SDGS) INTEGRATION

SDG 4 – Quality Education

The system enhances digital learning by identifying skill gaps in candidates and recommending relevant learning resources. It promotes continuous skill development through quizzes, assessments, and guided career roadmaps, enabling users to improve their knowledge and employability**.**

SDG 8 – Decent Work and Economic Growth

The platform improves employment opportunities by enabling accurate skill-based job matching. It reduces unemployment and underemployment by helping candidates find suitable jobs while assisting companies in hiring qualified individuals efficiently, thereby supporting economic growth.

SDG 9 – Industry, Innovation, and Infrastructure

The project leverages modern technologies such as AI-based skill analysis, matching algorithms, and cloud-based infrastructure to build an innovative recruitment system. It contributes to the development of intelligent digital hiring solutions.

SDG 11 – Sustainable Cities and Communities

By enabling efficient online recruitment and remote skill validation, the platform reduces the need for physical travel, supporting sustainable urban development and promoting digital accessibility.

SDG 12 – Responsible Consumption and Production

The system promotes efficient use of time and resources by preventing irrelevant job applications through accurate skill-job matching, reducing unnecessary effort for both candidates and recruiters.

SDG 16 – Peace, Justice, and Strong Institutions

The platform ensures transparency, fairness, and security in the hiring process through verified skill assessments, secure authentication mechanisms, and unbiased screening, strengthening trust in digital recruitment systems.

SDG	GOALS	Level	Integration
4	Quality Education	3	Enhances learning through skill gap analysis, quizzes, and career roadmap recommendations.
8	Decent Work and Economic Growth	3	Improves employment opportunities via accurate skill-job matching and efficient hiring.
9	Industry, Innovation, and Infrastructure	3	Uses AI, algorithms, and cloud infrastructure for innovative recruitment solutions.
11	Sustainable Cities and Communities	2	Supports remote hiring and reduces physical resource usage.
12	Responsible Consumption and Production	2	Minimizes resource wastage by preventing irrelevant job applications.
16	Peace, Justice, and Strong Institutions	3	Ensures fair, transparent, and secure recruitment processes.

LOW – 1, MEDIUM – 2, HIGH – 3

ABSTRACT

This project proposes the development of a web-based structure works best, utilizing HTML, CSS, and JavaScript at the front end for a seamless and friendly interface. While java is used to run the server to run the business logic, authentication, and APIs. To run the system, a H2 is used to store user profiles, job postings, skill assessments, application records. To step out of the current service of the competitors, the portal adds features like AI-based service for skills analysis and job recommendations. Candidate skills and assessment game results along with the target jobs are processed through a matching algorithm to produce compatibility scores. Timed quizzes and small task submissions as online skills verification tools are used to legitimize user skills. No-name hiring components are used in the first round of screening to eliminate the risk of bias.

A few more services would be real-time alerts using email/SMS APIs, secure authentication via JWT/OAuth and cloud deployment capabilities for higher scaling and availability. The platform also offers career roadmap features that help users identify skill gaps and recommends learning resources. The innovative job portal showcases the impact modern tools and intelligent services can have on increasing transparency, visibility and accuracy in recruitment, providing an inspired break from traditional job platforms.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
DB	Database
HTML	HyperText Markup Language
JWT	JSON Web Token
ML	Machine Learning
MVT	Model View Template
MySQL	My Structured Query Language
NLP	Natural Language Processing
OAuth	Open Authorization
OS	Operating System
REST API	Representational State Transfer Application Programming Interface
SMS	Short Message Service
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

Today's competitive labor market has resulted in many challenges that both job seekers and recruiters have to overcome while going through the recruitment process. The greatest challenge both parties face is that there is a disconnect between candidates' skill sets and the requirements of the job they applied for. Both sides make the task of recruiting difficult for themselves due to this disconnect. Candidates apply for jobs without fully understanding whether their skills meet the expectations of the position they are applying for so they are often rejected and feel as though they wasted their time and effort. At the same time, recruiters have a difficult time recruiting suitable candidates due to the sheer number of applications they receive and the lack of reliable methods to verify a candidate's skill set.

The traditional job portal (job board) process depends upon resumes and the use of keywords to filter out candidates for selection. Although the use of these systems can help manage applications they often do not accurately represent a candidate's actual skill set. A candidate's resume can have false and/or missing information and keyword based filtering will sometimes omit qualified candidates who have not used the required keywords. Therefore the recruitment process can be inefficient, time-consuming and error-prone.

The current recruitment system contains yet another major issue - insufficient validation of candidate's abilities via proper skill assessment. There are very few platforms in existence which provide means to validate candidate's skillset through a practical evaluation. As a result, recruiters are left with no choice but to depend primarily on the contents of resumes and the results of interview sessions to base hiring decisions upon. Unfortunately, many of these documents are not a true representation of the abilities that applicants possess and therefore will have a negative impact when determining an optimal hire for the organization. Further compounding this problem, both consciously and unconsciously, bias toward certain types of candidates will continue to hinder hiring decisions, leading to inequities in opportunity.

To address this significant barrier to effective recruitment, there has been the development of the "Skill Validation & Hiring Platform" (the "Platform"), a web-based smart solution for improving the recruitment process by introducing an intermediary between job seekers and recruiters. By utilizing modern web-based technology, the Platform integrates intelligent features into the process to help improve efficiencies and increase the accuracy of candidate evaluations and ultimately hiring decisions by focusing on the skill set possessed by each candidate as opposed to relying solely on a candidate's resume as a basis to evaluate them and make hiring decisions.

By providing a platform for users to register, create profiles, and manage their skill sets, candidates have the ability to upload their resumes to the system for analysis. This analysis assesses both the structure and relevance of the resume to calculate a quality score, which assists in improving the overall quality of a candidate's resume and maximizing the probability of them being selected. The system also offers online assessments to evaluate the knowledge of the candidate through quizzes and tasks related to different domains. By conducting these assessments, the skills demonstrated by the candidates are verified in an accurate and objective manner.

A primary objective of this proposed system is the intelligent job matching mechanism, which utilizes a matching algorithm to compare candidate skill sets, results from skill assessments and job requirements. The matching algorithm yields a compatibility score which allows recruiters to quickly and accurately determine the best matched candidates for open positions. In addition, the system generates personalized job recommendations for candidates based upon their skill set and performance, thereby increasing their chances of being matched with relevant employment opportunities.

The proposed system will also include a mechanism for anonymous hiring to eliminate bias during the early stages of the hiring process. Personal information such as a candidate's name and gender will be hidden, thereby allowing recruiters to base their evaluations of candidates solely upon their skill set and performance. By doing so, the system encourages fair hiring practices and provides all candidates with an equal opportunity to attain employment.

Moreover, the application provides a skills gap analysis feature that identifies any skills missing or lacking that would be necessary for a position. This also provides recommendations for improving those missing areas as well as specific learning resources to help candidates increase their skills in those areas. Both the applicant and the employer support this new form of learning and career growth and therefore the application acts as not only a recruitment application but as a skills development one.

In general, the new proposed application will change the current recruiting process by creating an application that has efficiency, transparency, and a heavy focus on skills ultimately enhancing the experience for BOTH job seekers and employers. By including skill verification, intelligent matching of candidates with jobs, and eliminating bias in the interview and hiring process, the proposed application will decrease the time spent in the recruitment process, greatly improve the precision of hiring, and create a much better usability for job seekers and employers.

1.1 OBJECTIVES

1. Secure User Management

To design and implement a secure system for user registration, login, and authentication, ensuring safe access for both candidates and recruiters.

2. Skill-Based Profile Management

To enable users to create and manage their profiles by adding relevant skills, qualifications, and experience for better representation.

3. Resume Analysis System

To develop an automated resume evaluation module that analyzes uploaded resumes and generates a quality score based on structure and relevance.

4. Skill Assessment Mechanism

To provide online quizzes and task-based evaluations that validate the practical knowledge and abilities of candidates.

5. Intelligent Job Matching

To implement a matching algorithm that compares candidate skills with job requirements and generates compatibility scores for accurate job recommendations.

6. Skill Gap Identification

To identify missing or weak skills required for specific job roles and provide suggestions for improvement.

7. Bias-Free Recruitment

To incorporate an anonymous hiring feature that hides personal details during initial screening to ensure fairness and reduce bias.

8. Enhanced Recruitment Efficiency

To improve the overall efficiency, transparency, and accuracy of the recruitment process for both job seekers and recruiters.

CHAPTER 2

LITERATURE SURVEY

1. A. Kumar, P. Singh, and R. Patel, "Intelligent job matching using machine learning techniques," International Journal of Computer Science and Engineering, vol. 11, no. 3, pp. 45–52, 2023.

A recent study done by Kumar, Singh and Patel (2023) explores how using machine learning could create a better system for matching jobs. Their system is simpler than other commonly used methods of matching people to jobs as it only uses keywords from resumes and job postings. Instead of using only keywords to match people to jobs they will use the skills, education and experience of each candidate in addition to those used by traditional methods.

Kumar et. al. will be applying machine learning algorithms to match candidates with their preferred types of jobs. Through this process they will then create a compatibility score for each candidate. As the authors discuss throughout the study, automation can minimize the amount of human interaction involved in hiring while providing faster and more accurate results for decision making purposes.

Moreover, the research discusses challenges such as dependencies on data, model training and the complexity of building the model. In spite of these limitations, it is concluded that intelligent job matching systems are an enhancement to the accuracy and efficiency of recruitment processes. This research helps to support the proposed system in that it expands upon these concepts by adding features for assessing skills, analyzing resumes and identifying skill gaps.[1]

2. S. Sharma and R. Verma, "AI-based resume screening and skill assessment system," International Journal of Advanced Research in Computer Science, vol. 9, no. 2, pp. 120–126, 2020..

S. Sharma and R. Verma (2020) explored the potential uses for AI technologies in streamlining both assessment of resumes and verification of skills. Historically, when Recruiters have evaluated candidates for recruitment, they would manually review each applicant's Resumes. This process takes a substantial amount of time and is often subjective. Furthermore, candidates' resumes may not reflect a true representation of their skillset; consequently, some of the most qualified candidates might be overlooked.

The researchers proposed an alternative to evaluating applicants through the use of an AI-based resume review/evaluation system. Ultimately, their AI program would leverage Natural

Language Processing (NLP) tools to extract critical information from each candidate's Resume (i.e., Skills, Qualifications, Work Experience, Certifications, etc.). After this extracted data was captured from each Resume, the Researchers would analyse it and identify whether or not each candidate would fit the specific Job Requirements outlined by the employer. Through automating the process of evaluating resumes, the AI resume evaluation system will significantly reduce the human resources used in evaluating resumes while also increasing consistency in evaluating resumes. In addition, using an AI-based resume evaluation system will ensure all critical data within the resume of potential candidates will be saved.

The system also helps to streamline the hiring process by allowing recruiters to devote more time to identifying the best candidates rather than manually filtering a large number of applications. Although the study does note some limitations, for example, difficulties in parsing resumes with various formats and ensuring fairness and impartiality in automated decision-making, it has also shown conclusively that automated resume screening and skill validation systems will substantially improve the recruitment process by providing a speedier, more accurate and reliable mechanism for assessing candidates. This study establishes the necessity for the development of automated resume evaluation and skill validation as components of any contemporary recruitment platform.[2]

3. J. Lee, "Analysis of traditional job portals and recommendation systems," Journal of Information Systems, vol. 7, no. 1, pp. 30–35, 2018.

The research of J. Lee (2018) discusses traditional job boards/recruitment websites/recruiting technologies and their recommendation capabilities, with emphasis on the weaknesses of utilizing these technologies in today's recruiting environment. Traditional job boards primarily use keyword filtering and manual selection processes as methods for matching candidates with positions. Job boards provide a basic platform for job seeking; however, job boards lack the necessary intelligence to appropriately match candidates with positions.

The study identifies as one of the primary challenges in recruiting, a lack of appropriate validation of skills. Candidates frequently apply for jobs using inaccurate or incomplete information regarding their qualifications. As a result, it is very common for candidates to be matched with jobs for which they do not have the requisite abilities. Recruiters also struggle to identify the right candidates because the tools currently available to them do not have the same level of reliability in evaluating candidate qualifications. As a result, recruitment processes require significant time and effort to identify suitable candidates.

Recommendation systems currently employed by traditional job boards generally rely on relatively simple filtering methods (i.e. keyword matching, selection based on basic user preferences, etc.). These filtering methods do not take into consideration the actual competency and/or capability of job seekers. As a result of this lack of assessment, recommendation systems frequently fail to provide job seekers with relevant and applicable job opportunities.

Consequently, job seekers may submit applications to jobs that are not aligned with their abilities and experience, and therefore many potential job opportunities for which they would otherwise qualify go unfilled.[3]

CHAPTER 3
PROPOSED WORK

3.1 Problem Statement

Currently, both job seekers and employers face a number of challenges within the recruiting process that make it difficult for them to find the best fit for their roles and candidates alike. Job seekers may apply for jobs without knowing how their experience fits into the requirements for the position, resulting in numerous rejections and low levels of confidence for candidates. Employers also face challenges associated with identifying potential hires, given that most applicant tracking systems process thousands of resumes without any verification of the candidate's actual skills. In addition, poorly constructed resumes and irrelevant applications make it difficult for employers to filter applications down to the best candidates. One of the significant issues still faced by employers in recruiting is the lack of intelligent job matching, which leads to an inadequate match between the skill set of the candidate and the job title. Furthermore, traditional applicant tracking systems do not provide applicants with any insights such as skills gaps or recommendations for skill-building, which affect candidates' ability to gain employment. These limitations demonstrate the pressing need for an advanced recruitment solution that accurately verifies candidates' skills, provides better job matches and provides job seekers with real guidance on how to enhance their employment prospects.

3.2 Proposed Methodology

The current design of the proposed system implements a systematic approach to validate skills efficiently and align candidates with job opportunities. The process begins with the candidate interacting with the system by registering and creating a profile on the platform and uploading a resume. The system collects data from each resume uploaded by the candidate before preprocessing it to extract relevant information concerning their skills, education and experience. The resume is then assessed based on its structure, relevance, and content by analysing the information collected; a score will be produced for each resume based on its level of quality.

Once candidates who have successfully matched their skills and knowledge to the prospective employer's requirements, they will be presented with skill gap analysis, which will show a candidate's missing or weak skills for the prospective employer's requirements and suggest areas for improvement. The system incorporates an anonymous screening process so that personal information is not viewable to prospective employers during the initial candidate assessment process; therefore, a more level recruitment playing field has been established. Lastly, the system has a notification module that keeps users informed of their application for employment status, skill assessment results and suggested improvements via real-time notifications.

3.3 System Architecture

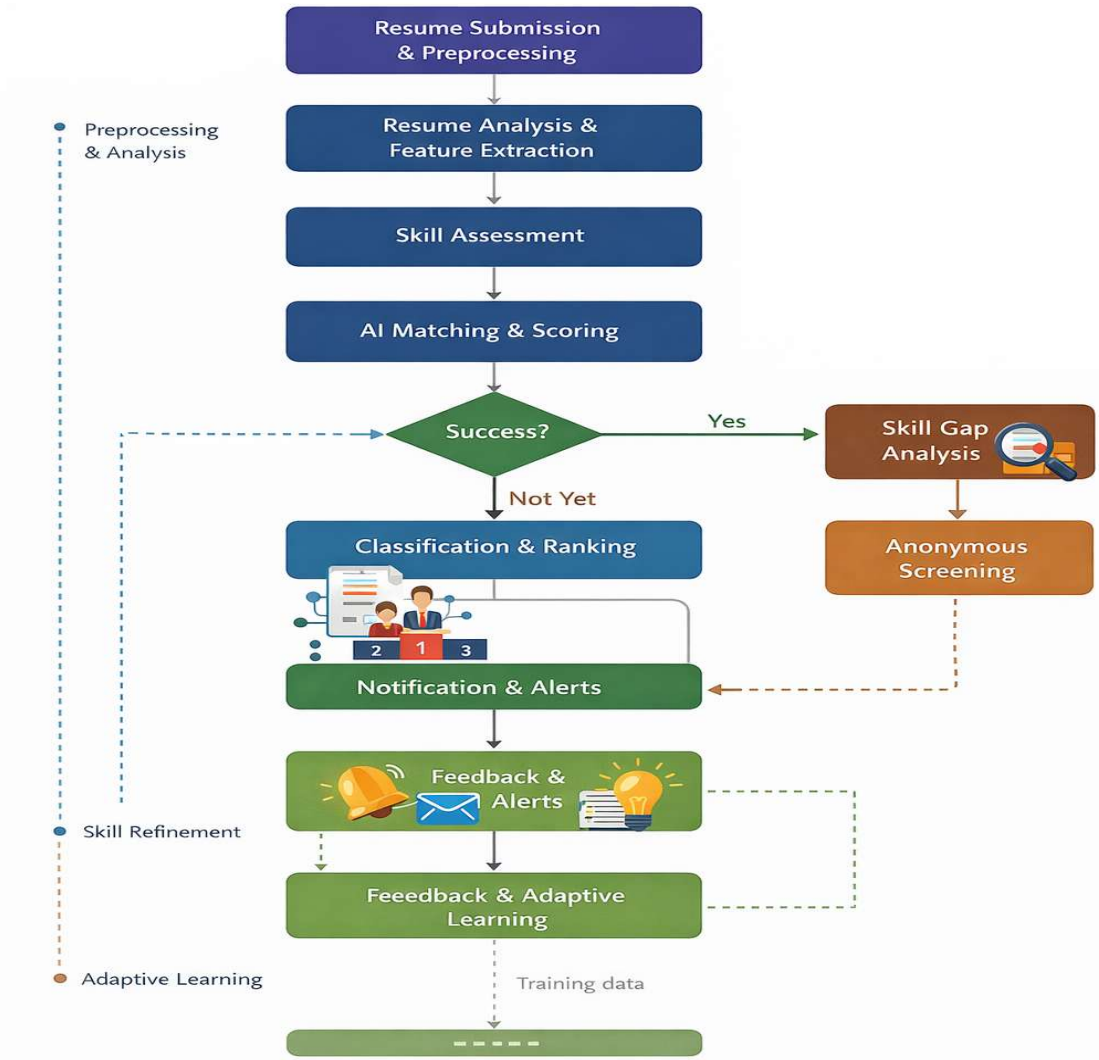


Fig 3.1: Flowchart of the SKILL VALIDATION AND HIRING PLATFORM proposed methodology depicting the end-to-end hybrid phishing detection and secure link inspection pipeline

The proposed system follows a sequential flow of modules, where each stage processes data and passes it to the next stage for intelligent decision-making.

1. User Interface Module

The process begins with user interaction through a web-based interface. Candidates can register, log in, upload resumes, and attend assessments, while recruiters can post job requirements and view candidate profiles.

2. Resume Submission and Preprocessing Module

Once the candidate uploads a resume, the system collects and preprocesses the data. It extracts relevant information such as skills, education, and experience, and removes inconsistencies to prepare structured data for analysis.

3. Resume Analysis and Feature Extraction Module

In this stage, the system evaluates the resume based on structure, keyword relevance, and completeness. Important features are extracted, and a resume score is generated to measure its quality**.**

4. Skill Assessment Module

Candidates undergo online quizzes and task-based evaluations to validate their practical knowledge. The results are recorded and used as a reliable measure of skill level.

5. AI-Based Matching and Scoring Module

The system compares candidate skills and assessment results with job requirements using an intelligent matching algorithm. A compatibility score is generated, indicating how well the candidate fits the job role.

6. Classification and Ranking Module

Based on the compatibility score, candidates are classified and ranked. Recruiters can easily identify the most suitable candidates from the ranked list.

7. Skill Gap Analysis Module

The system identifies missing or weak skills required for a particular job role. It provides recommendations to help candidates improve their qualifications.

8. Anonymous Screening Module

To ensure fairness, personal details such as name and gender are hidden during the initial screening process, reducing bias in recruitment.

9. Notification and Alert Module

The system sends real-time notifications regarding job applications, assessment results, and recommendations through email or SMS, keeping users informed.

10. Feedback and Learning Module

User feedback and system data are collected to improve performance. The matching algorithm is refined over time, enabling adaptive learning and better accuracy.

3.4 Modules

3.4.1 User Module

The User Module is designed to provide a seamless and interactive experience for job seekers using the platform. It begins with user registration and secure login, where candidates create accounts and authenticate themselves using secure mechanisms. Once logged in, users can create and manage their personal profiles by adding essential details such as education, experience, and technical skills.

The module allows users to continuously update their skills to reflect their current capabilities. Candidates can upload their resumes, which are then analyzed by the system to evaluate quality and relevance. A key feature of this module is the ability to participate in skill assessments and quizzes, which help validate the user's practical knowledge and competencies.

Based on the user's profile and assessment performance, the system provides personalized job recommendations, ensuring better alignment between candidate skills and job roles. Additionally, the platform identifies skill gaps by comparing user skills with job requirements, helping candidates understand areas for improvement. To support career growth, users can access learning resources and recommendations that guide them in enhancing their skills and improving employability.

3.4.2 Company Module

The Company Module is designed for recruiters and organizations to efficiently manage the hiring process. Companies can register and securely authenticate themselves to access the platform. Once logged in, they can post job openings by specifying required skills, job descriptions, and eligibility criteria.

Recruiters can view candidate profiles and resumes, which include detailed information about skills, qualifications, and assessment results. The system provides filtering mechanisms based on skill matching and compatibility scores, enabling companies to identify suitable candidates quickly.

Another important feature is the ability to review assessment results, which gives recruiters insights into the practical abilities of applicants. Companies can shortlist candidates based on their performance and manage job applications effectively within the system. Furthermore, the module supports communication with selected candidates, allowing recruiters to send interview invitations or updates, thereby streamlining the hiring process.

3.4.3 Admin Module

The Admin Module plays a crucial role in managing and maintaining the overall system. Administrators have the authority to manage user and company accounts, ensuring that only authorized users can access the platform. They monitor system activities and platform usage to maintain smooth operation and detect any irregularities.

The admin is responsible for managing job postings and maintaining platform data, ensuring that all information is accurate and up to date. Security is a major focus of this module, where administrators enforce access control mechanisms and protect user data from unauthorized access.

Additionally, the admin maintains the database and ensures optimal system performance by handling server-related tasks and data management processes. The module also includes reporting and analytics features, which generate insights about user activity, job trends, and system performance. These reports help in decision-making and continuous improvement of the platform.

3.5 Hardware Requirements

1. **Processor**: Intel Core i3 or higher
2. **RAM**: Minimum 4 GB (8 GB recommended)
3. **Storage**: Minimum 500 GB HDD / SSD
4. **Computer/Laptop**: System capable of running Django and development tools
5. **Input Devices**: Keyboard and Mouse
6. **Output Device**: Monitor
7. **Internet**: Stable Internet Connection

These hardware components are required to develop, run, and test Skill Validation and Hiring Platform website effectively

3.6. SOFTWARE REQUIREMENTS

1. **Operating System**: Windows 10 / Linux / macOS
2. **Frontend Technologies**: HTML, CSS, JavaScript
3. **Backend Technology**: Java Springboot
4. **Database**: H2
5. **Web Browser**: Google Chrome / Mozilla Firefox
6. Development Tools: VS Code / PyCharm
7. **Server**: springboot Development Server
8. **APIs**: REST API for communication
9. **Version Control**: Git

These software components are required to develop, run, and test Skill Validation and Hiring Platform website effectively

CHAPTER 4

SOFTWARE TESTING

4.1 UNIT TESTING

Unit testing focuses on verifying individual modules or components of the system independently to ensure each part functions correctly.

1. User Authentication Module:

Test user registration and login functionality to ensure correct validation of credentials and secure access to the system.

2. Resume Analysis Module:

Verify that the system correctly extracts data from uploaded resumes and generates an accurate resume score based on structure and relevance.

3. Skill Assessment Module:

Ensure that quiz questions are displayed correctly, user responses are recorded properly, and scores are calculated accurately.

4. Job Matching Module:

Test whether the system correctly compares candidate skills with job requirements and generates an appropriate compatibility score.

5. Input Validation:

Check the system's ability to handle invalid or incomplete inputs and ensure proper error messages are displayed to the user.

4.2 SYSTEM TESTING

System testing evaluates the complete system to ensure that all modules work together correctly and meet the required functionality**.**

1.User Interface Testing:

Verify that all pages such as home, login, profile, and job listings load correctly and provide a smooth user experience.

2. Backend Processing:

Ensure that the Django backend processes user requests properly and communicates effectively with the database and other modules.

3. Database Integration:

Check whether user data, resumes, job details, and assessment results are stored and retrieved correctly from the MySQL database.

4. API Communication:

Verify that REST APIs function correctly and enable smooth data exchange between frontend and backend.

5. End-to-End Workflow:

Test the complete flow from user registration → skill assessment → job matching → recommendation to ensure the system works as expected.

4.3 ALGORITHM TESTING

Algorithm testing ensures the correctness and efficiency of the algorithms used in the system.

1. Resume Analysis Algorithm:

Validate that the algorithm accurately extracts features and generates a correct resume quality score.

2. Skill Assessment Algorithm:

Ensure that the scoring logic correctly evaluates user performance based on quiz responses.

3. Job Matching Algorithm:

Verify that the system accurately compares candidate skills with job requirements and produces reliable compatibility scores.

4. Recommendation Algorithm:

Check whether the system provides relevant job and learning recommendations based on user profile and performance.

5. Performance Testing:

Ensure that all algorithms execute efficiently without delays, even with large datasets.

CHAPTER 5

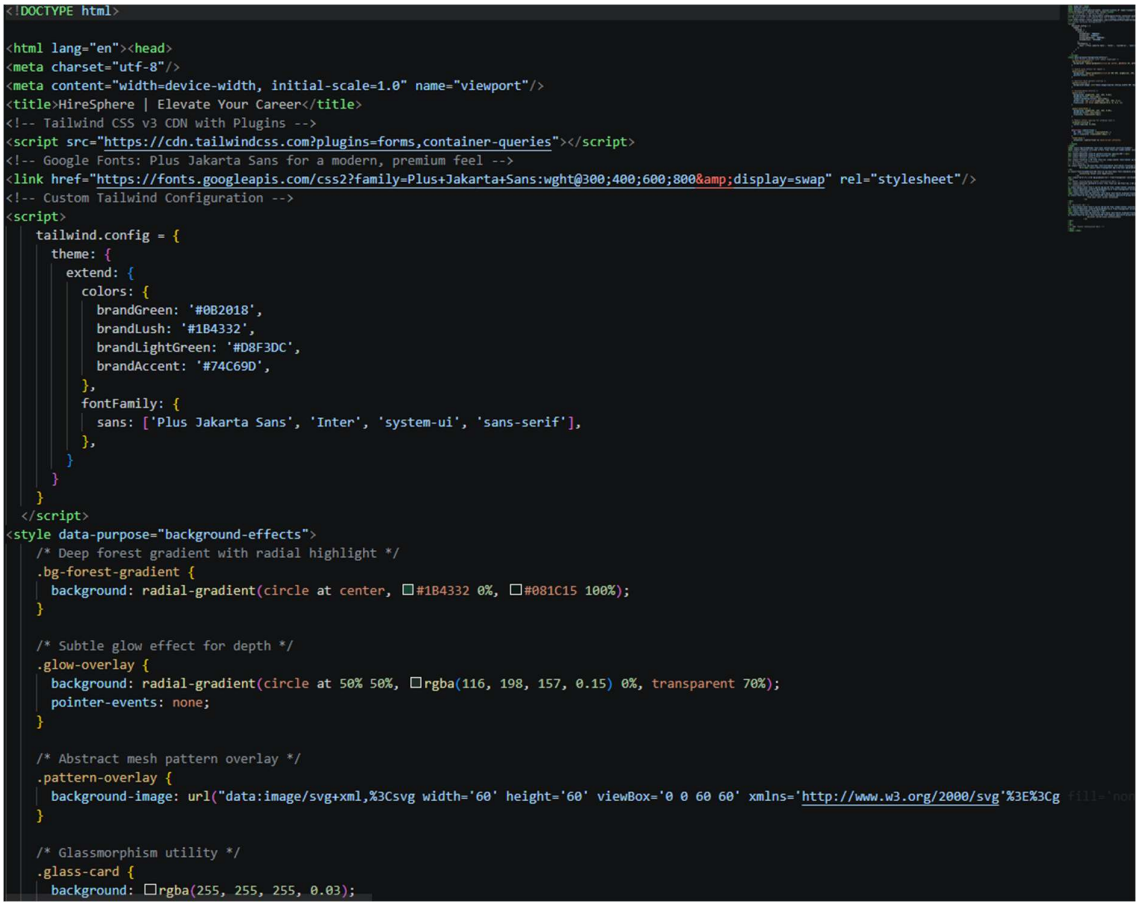
SOURCE CODE

A web-based platform is built using the Python programming language through the Spring boot framework on the back end and HTML, CSS, and JavaScript on the front end. The code base within this application is organized into modules that correspond to user administration, job processing, assessments, and recommendations.

The database structures are defined within Spring boot models.py file; the business logic and other application logic are defined in views.py; and the application's routing is defined in urls.py. In addition to these elements, the front end utilizes the Spring boot template system to render dynamic content and provide a user-friendly interface.

Frontend

1. Login page



```
<!DOCTYPE html>

<html lang="en"><head>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1.0" name="viewport"/>
<title>HireSphere | Elevate Your Career</title>
<!-- Tailwind CSS v3 CDN with Plugins -->
<script src="https://cdn.tailwindcss.com?plugins=forms,container-queries"></script>
<!-- Google Fonts: Plus Jakarta Sans for a modern, premium feel -->
<link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@300;400;600;800&amp;display=swap" rel="stylesheet"/>
<!-- Custom Tailwind Configuration -->
<script>
    tailwind.config = {
      theme: {
        extend: {
          colors: {
            brandGreen: '#0B2018',
            brandLush: '#1B4332',
            brandLightGreen: '#D8F3DC',
            brandAccent: '#74C69D',
          },
          fontFamily: {
            sans: ['Plus Jakarta Sans', 'Inter', 'system-ui', 'sans-serif'],
          },
        }
      }
    }
  </script>
<style data-purpose="background-effects">
    /* Deep forest gradient with radial highlight */
    .bg-forest-gradient {
      background: radial-gradient(circle at center, #1B4332 0%, #081C15 100%);
    }

    /* Subtle glow effect for depth */
    .glow-overlay {
      background: radial-gradient(circle at 50% 50%, rgba(116, 198, 157, 0.15) 0%, transparent 70%);
      pointer-events: none;
    }

    /* Abstract mesh pattern overlay */
    .pattern-overlay {
      background-image: url("data:image/svg+xml,%3Csvg width='60' height='60' viewBox='0 0 60 60' xmlns='http://www.w3.org/2000/svg'%3E%3Cg
    }

    /* Glassmorphism utility */
    .glass-card {
      background: rgba(255, 255, 255, 0.03);
```

```
<script>
      theme: {
        extend: {
          },
        }
      }
    }
  </script>
<style data-purpose="background-effects">
    /* Deep forest gradient with radial highlight */
    .bg-forest-gradient {
      background: radial-gradient(circle at center, #1B4332 0%, #081C15 100%);
    }

    /* Subtle glow effect for depth */
    .glow-overlay {
      background: radial-gradient(circle at 50% 50%, rgba(116, 198, 157, 0.15) 0%, transparent 70%);
      pointer-events: none;
    }

    /* Abstract mesh pattern overlay */
    .pattern-overlay {
      background-image: url("data:image/svg+xml,%3Csvg width='60' height='60' viewBox='0 0 60 60' xmlns='http://www.w3.org/2000/svg'%3E%3Cg
    }

    /* Glassmorphism utility */
    .glass-card {
      background: rgba(255, 255, 255, 0.03);
      backdrop-filter: blur(16px);
      -webkit-backdrop-filter: blur(16px);
      border-top: 1px solid rgba(255, 255, 255, 0.1);
      transition: all 0.4s cubic-bezier(0.4, 0, 0.2, 1);
    }

    .glass-card:hover {
      background: rgba(255, 255, 255, 0.08);
      backdrop-filter: blur(24px);
      transform: translateY(-4px);
    }

    /* Modern letter spacing for premium look */
    .premium-tracking {
      letter-spacing: 0.15em;
    }

    @keyframes subtle-float {
      0%, 100% { transform: translateY(0); }
```

```
    @keyframes subtle-float {
      50% { transform: translateY(-10px); }
    }
    .float-anim {
      animation: subtle-float 6s ease-in-out infinite;
    }
  </style>
</head>
<body class="bg-brandGreen font-sans antialiased overflow-hidden">
<main class="relative h-screen w-full flex flex-col items-center justify-center bg-forest-gradient" data-purpose="landing-hero">
<!-- Visual Overlays -->
<div class="absolute inset-0 pattern-overlay opacity-50"></div>
<div class="absolute inset-0 glow-overlay"></div>
<!-- Center Content: Branding -->
<div class="relative z-10 flex flex-col items-center text-center px-4">
<!-- HireSphere Logo Placeholder -->
<!-- Main Heading -->
<h1 class="text-7xl md:text-9xl font-extrabold text-white tracking-tighter mb-4">
            Hire<span class="text-transparent bg-clip-text bg-gradient-to-r from-brandAccent to-emerald-300">Sphere</span>
</h1>
<p class="text-brandAccent/80 text-sm md:text-base font-semibold premium-tracking uppercase">
            Connecting Talent with Opportunity
        </p>
<div class="mt-8 h-1 w-24 bg-gradient-to-r from-transparent via-brandAccent to-transparent opacity-50"></div>
</div>
<!-- BEGIN: Glassmorphism Footer Interaction Bars -->
<div class="absolute bottom-0 w-full flex flex-col md:flex-row z-20 gap-px bg-white/5" data-purpose="user-entry-points">
<!-- Candidate Bar -->
<a class="glass-card flex-1 py-12 md:py-16 flex items-center justify-center group relative overflow-hidden" href="p1home.html">
<div class="absolute inset-0 bg-gradient-to-b from-transparent to-brandAccent/5 opacity-0 group-hover:opacity-100 transition-opacity durati
<div class="text-center relative z-10">
<span class="text-2xl md:text-3xl font-bold text-white premium-tracking uppercase group-hover:text-brandAccent transition-colors">Candidate
<p class="text-sm mt-3 text-white/50 font-medium opacity-0 group-hover:opacity-100 transition-all duration-300 transform translate-y-2 grou
                Find your next career milestone
            </p>
</div>
</a>
<!-- Recruiter Bar -->
<a class="glass-card flex-1 py-12 md:py-16 flex items-center justify-center group relative overflow-hidden" href="p25re.html">
<div class="absolute inset-0 bg-gradient-to-b from-transparent to-brandAccent/5 opacity-0 group-hover:opacity-100 transition-opacity durati
<div class="text-center relative z-10">
<span class="text-2xl md:text-3xl font-bold text-white premium-tracking uppercase group-hover:text-brandAccent transition-colors">Recruiter
<p class="text-sm mt-3 text-white/50 font-medium opacity-0 group-hover:opacity-100 transition-all duration-300 transform translate-y-2 grou
                Discover world-class professionals
            </p>
</div>
```

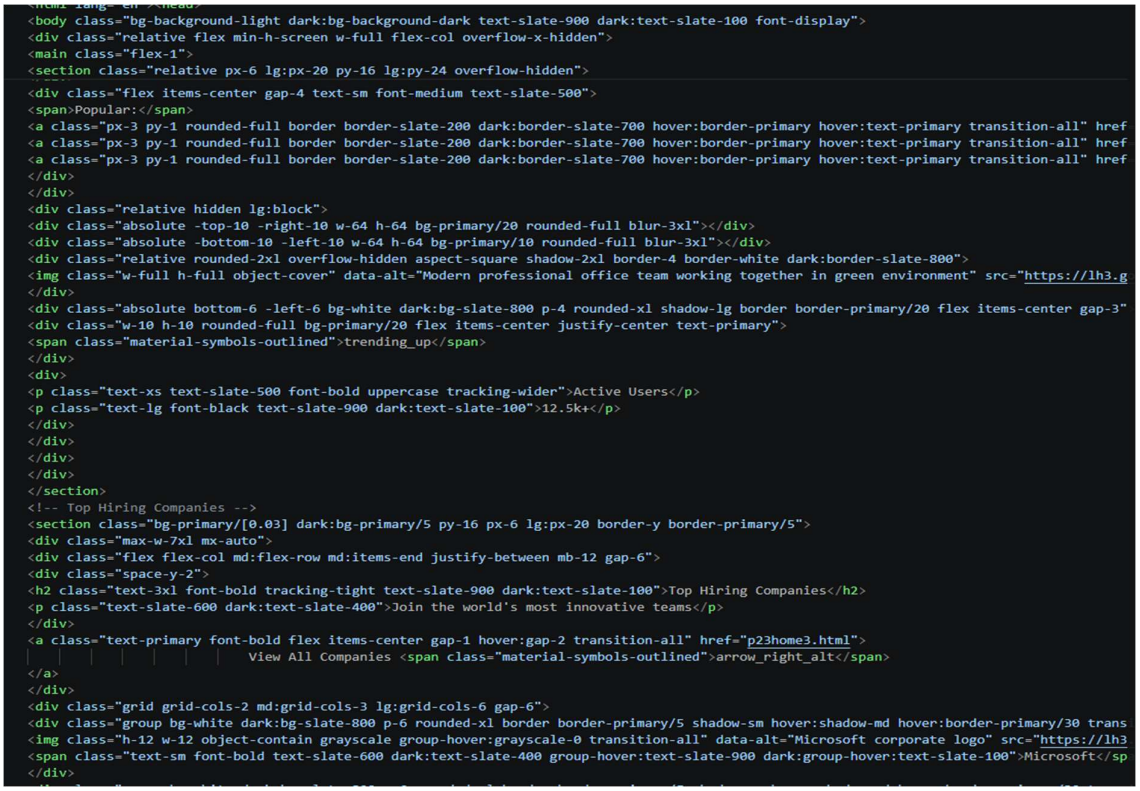
2. Home page

```
<!DOCTYPE html>

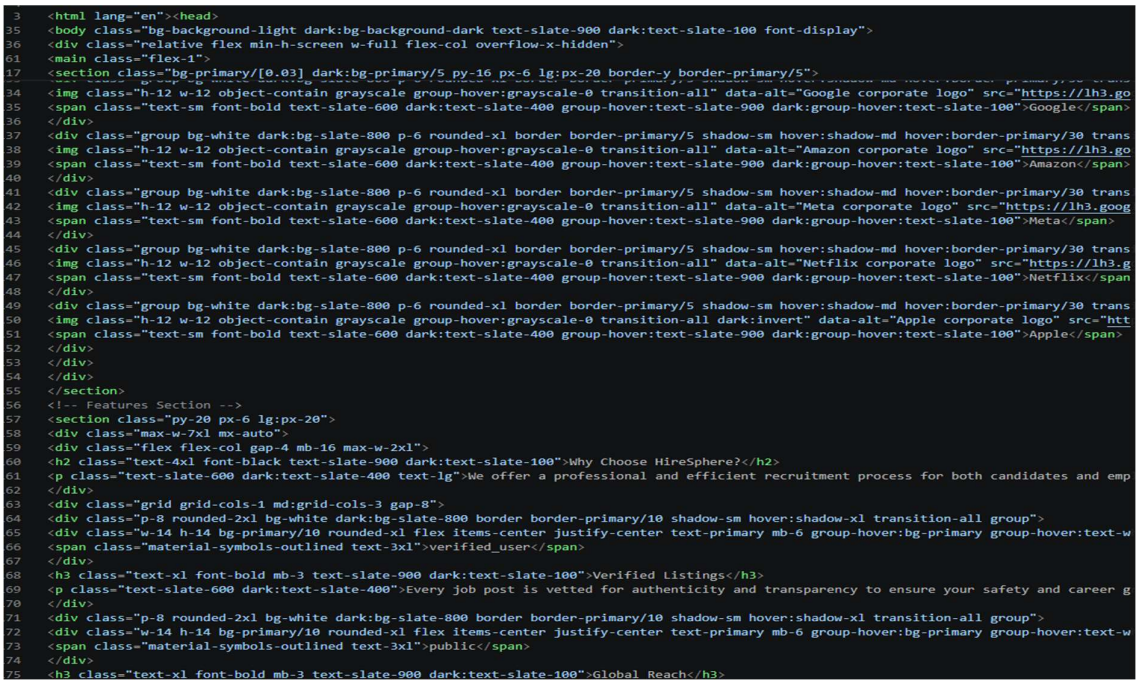
<html lang="en"><head>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1.0" name="viewport"/>
<title>HireSphere - Find Your Dream Job</title>
<script src="https://cdn.tailwindcss.com?plugins=forms,container-queries"></script>
<link href="https://fonts.googleapis.com/css2?family=Manrope:wght@200..800&amp;display=swap" rel="stylesheet"/>
<link href="https://fonts.googleapis.com/css2?family=Material+Symbols+Outlined:wght,FILL@100..700,0..1&amp;display=swap" rel="stylesheet"/>
<link href="https://fonts.googleapis.com/css2?family=Material+Symbols+Outlined:wght,FILL@100..700,0..1&amp;display=swap" rel="stylesheet"/>
<script id="tailwind-config">
        tailwind.config = {
            darkMode: "class",
            theme: {
                extend: {
                    colors: {
                        "primary": "#11d421",
                        "background-light": "#f6f8f6",
                        "background-dark": "#102212",
                    },
                    fontFamily: {
                        "display": ["Manrope", "sans-serif"]
                    },
                    borderRadius: {"DEFAULT": "0.25rem", "lg": "0.5rem", "xl": "0.75rem", "full": "9999px"},
                },
            },
        }
    </script>
<style>
        body {
            font-family: 'Manrope', sans-serif;
        }
    </style>
</head>
<body class="bg-background-light dark:bg-background-dark text-slate-900 dark:text-slate-100 font-display">
<div class="relative flex min-h-screen w-full flex-col overflow-x-hidden">
<!-- Navigation Bar -->
<header class="sticky top-0 z-50 w-full border-b border-primary/10 bg-background-light/80 dark:bg-background-dark/80 backdrop-blur-md px-6
<div class="max-w-7xl mx-auto flex items-center justify-between">
<div class="flex items-center gap-2 text-primary">
<span class="material-symbols-outlined text-3xl font-bold">corporate_fare</span>
<a h2 class="text-xl font-extrabold tracking-tight text-slate-900 dark:text-slate-100" href="p0landing.html">HireSphere</h2></a>
</div>
<div class="flex items-center gap-8">
<nav class="hidden md:flex items-center gap-8">
<a class="text-sm font-semibold hover:text-primary transition-colors" href="p23home3.html">Companies</a>
<a class="text-sm font-semibold hover:text-primary transition-colors" href="p2aboutas.html">About Us</a>
</nav>
```

```
    <html lang="en"><head>
    <body class="bg-background-light dark:bg-background-dark text-slate-900 dark:text-slate-100 font-display">
    <div class="relative flex min-h-screen w-full flex-col overflow-x-hidden">
    <header class="sticky top-0 z-50 w-full border-b border-primary/10 bg-background-light/80 dark:bg-background-dark/80 backdrop-blur-md px-6
    <div class="max-w-7xl mx-auto flex items-center justify-between">
    </nav>
    <div class="flex items-center gap-3 pl-8 border-l border-slate-200 dark:border-slate-700">
    <button class="flex items-center gap-2 group">

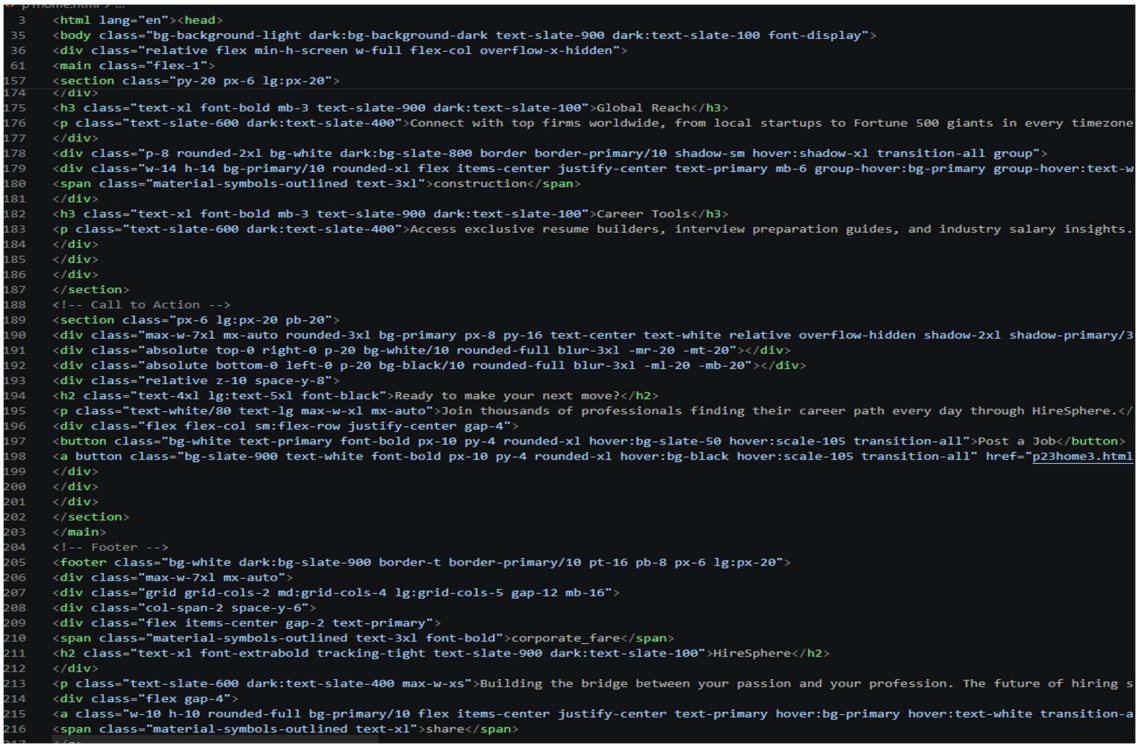
    <a button class="px-5 py-2 text-sm font-bold bg-primary/10 text-slate-900 dark:text-slate-100 rounded-lg hover:bg-primary/20 transition-all
                        Login
                    </button></a>
    <a button class="px-5 py-2 text-sm font-bold bg-primary text-white rounded-lg shadow-lg shadow-primary/20 hover:scale-105 transition-all" h
                        Sign Up
                    </button></a>
    </div>
    </div>
    </header>
    <main class="flex-1">
    <!-- Hero Section -->
    <section class="relative px-6 lg:px-20 py-16 lg:py-24 overflow-hidden">
    <div class="max-w-7xl mx-auto grid lg:grid-cols-2 gap-12 items-center">
    <div class="flex flex-col gap-8">
    <div class="space-y-4">
    <span class="inline-block px-3 py-1 text-xs font-bold tracking-widest uppercase bg-primary/20 text-primary rounded-full">
                            World's #1 Job Platform
                        </span>
    <h1 class="text-5xl lg:text-7xl font-black leading-tight tracking-tighter text-slate-900 dark:text-slate-100">
                            Find the <span class="text-primary">perfect job</span> for you
                        </h1>
    <p class="text-lg text-slate-600 dark:text-slate-400 max-w-lg">
                            Connecting talent with opportunity across the globe. We streamline the recruitment process to help you land
                        </p>
    </div>
    <!-- Search Bar -->
    <div class="w-full max-w-2xl bg-white dark:bg-slate-800 p-2 rounded-xl shadow-xl border border-primary/10 flex flex-col md:flex-row gap-2">
    <div class="flex-1 flex items-center gap-3 px-4 py-3 border-b md:border-b-0 md:border-r border-slate-100 dark:border-slate-700">
    <span class="material-symbols-outlined text-primary">search</span>
    <input class="w-full bg-transparent border-none focus:ring-0 text-sm font-medium placeholder-slate-400" placeholder="Job Title, Keywords" t
    </div>
    <div class="flex-1 flex items-center gap-3 px-4 py-3">
    <span class="material-symbols-outlined text-primary">location_on</span>
    <input class="w-full bg-transparent border-none focus:ring-0 text-sm font-medium placeholder-slate-400" placeholder="Location" type="text"/
    </div>
    <button class="bg-primary hover:bg-primary/90 text-white font-bold py-3 px-8 rounded-lg transition-all flex items-center justify-center gap
    <span>Search</span>
    </button>
    </div>
```



```
<body class="bg-background-light dark:bg-background-dark text-slate-900 dark:text-slate-100 font-display">
<div class="relative flex min-h-screen w-full flex-col overflow-x-hidden">
<main class="flex-1">
<section class="relative px-6 lg:px-20 py-16 lg:py-24 overflow-hidden">
<div class="flex items-center gap-4 text-sm font-medium text-slate-500">
<span>Popular:</span>
<a class="px-3 py-1 rounded-full border border-slate-200 dark:border-slate-700 hover:border-primary hover:text-primary transition-all" href
<a class="px-3 py-1 rounded-full border border-slate-200 dark:border-slate-700 hover:border-primary hover:text-primary transition-all" href
<a class="px-3 py-1 rounded-full border border-slate-200 dark:border-slate-700 hover:border-primary hover:text-primary transition-all" href
</div>
</div>
<div class="relative hidden lg:block">
<div class="absolute -top-10 -right-10 w-64 h-64 bg-primary/20 rounded-full blur-3xl"></div>
<div class="absolute -bottom-10 -left-10 w-64 h-64 bg-primary/10 rounded-full blur-3xl"></div>
<div class="relative rounded-2xl overflow-hidden aspect-square shadow-2xl border-4 border-white dark:border-slate-800">
<img class="w-full h-full object-cover" data-alt="Modern professional office team working together in green environment" src="https://lh3.g
</div>
<div class="absolute bottom-6 -left-6 bg-white dark:bg-slate-800 p-4 rounded-xl shadow-lg border border-primary/20 flex items-center gap-3"
<div class="w-10 h-10 rounded-full bg-primary/20 flex items-center justify-center text-primary">
<span class="material-symbols-outlined">trending_up</span>
</div>
<div>
<p class="text-xs text-slate-500 font-bold uppercase tracking-wider">Active Users</p>
<p class="text-lg font-black text-slate-900 dark:text-slate-100">12.5k+</p>
</div>
</div>
</div>
</div>
</section>
<!-- Top Hiring Companies -->
<section class="bg-primary/[0.03] dark:bg-primary/5 py-16 px-6 lg:px-20 border-y border-primary/5">
<div class="max-w-7xl mx-auto">
<div class="flex flex-col md:flex-row md:items-end justify-between mb-12 gap-6">
<div class="space-y-2">
<h2 class="text-3xl font-bold tracking-tight text-slate-900 dark:text-slate-100">Top Hiring Companies</h2>
<p class="text-slate-600 dark:text-slate-400">Join the world's most innovative teams</p>
</div>
<a class="text-primary font-bold flex items-center gap-1 hover:gap-2 transition-all" href="p23home3.html">
                    View All Companies <span class="material-symbols-outlined">arrow_right_alt</span>
</a>
</div>
<div class="grid grid-cols-2 md:grid-cols-3 lg:grid-cols-6 gap-6">
<div class="group bg-white dark:bg-slate-800 p-6 rounded-xl border border-primary/5 shadow-sm hover:shadow-md hover:border-primary/30 trans
<img class="h-12 w-12 object-contain grayscale group-hover:grayscale-0 transition-all" data-alt="Microsoft corporate logo" src="https://lh3
<span class="text-sm font-bold text-slate-600 dark:text-slate-400 group-hover:text-slate-900 dark:group-hover:text-slate-100">Microsoft</sp
</div>
```



```
<html lang="en"><head>
<body class="bg-background-light dark:bg-background-dark text-slate-900 dark:text-slate-100 font-display">
<div class="relative flex min-h-screen w-full flex-col overflow-x-hidden">
<main class="flex-1">
<section class="bg-primary/[0.03] dark:bg-primary/5 py-16 px-6 lg:px-20 border-y border-primary/5">
<img class="h-12 w-12 object-contain grayscale group-hover:grayscale-0 transition-all" data-alt="Google corporate logo" src="https://lh3.go
<span class="text-sm font-bold text-slate-600 dark:text-slate-400 group-hover:text-slate-900 dark:group-hover:text-slate-100">Google</span>
</div>
<div class="group bg-white dark:bg-slate-800 p-6 rounded-xl border border-primary/5 shadow-sm hover:shadow-md hover:border-primary/30 trans
<img class="h-12 w-12 object-contain grayscale group-hover:grayscale-0 transition-all" data-alt="Amazon corporate logo" src="https://lh3.go
<span class="text-sm font-bold text-slate-600 dark:text-slate-400 group-hover:text-slate-900 dark:group-hover:text-slate-100">Amazon</span>
</div>
<div class="group bg-white dark:bg-slate-800 p-6 rounded-xl border border-primary/5 shadow-sm hover:shadow-md hover:border-primary/30 trans
<img class="h-12 w-12 object-contain grayscale group-hover:grayscale-0 transition-all" data-alt="Meta corporate logo" src="https://lh3.goog
<span class="text-sm font-bold text-slate-600 dark:text-slate-400 group-hover:text-slate-900 dark:group-hover:text-slate-100">Meta</span>
</div>
<div class="group bg-white dark:bg-slate-800 p-6 rounded-xl border border-primary/5 shadow-sm hover:shadow-md hover:border-primary/30 trans
<img class="h-12 w-12 object-contain grayscale group-hover:grayscale-0 transition-all" data-alt="Netflix corporate logo" src="https://lh3.g
<span class="text-sm font-bold text-slate-600 dark:text-slate-400 group-hover:text-slate-900 dark:group-hover:text-slate-100">Netflix</span
</div>
<div class="group bg-white dark:bg-slate-800 p-6 rounded-xl border border-primary/5 shadow-sm hover:shadow-md hover:border-primary/30 trans
<img class="h-12 w-12 object-contain grayscale group-hover:grayscale-0 transition-all dark:invert" data-alt="Apple corporate logo" src="htt
<span class="text-sm font-bold text-slate-600 dark:text-slate-400 group-hover:text-slate-900 dark:group-hover:text-slate-100">Apple</span>
</div>
</div>
</div>
</section>
<!-- Features Section -->
<section class="py-20 px-6 lg:px-20">
<div class="max-w-7xl mx-auto">
<div class="flex flex-col gap-4 mb-16 max-w-2xl">
<h2 class="text-4xl font-black leading-tight text-slate-900 dark:text-slate-100">Why Choose HireSphere?</h2>
<p class="text-slate-600 dark:text-slate-400 text-lg">We offer a professional and efficient recruitment process for both candidates and emp
</div>
<div class="grid grid-cols-1 md:grid-cols-3 gap-8">
<div class="p-8 rounded-2xl bg-white dark:bg-slate-800 border border-primary/10 shadow-sm hover:shadow-xl transition-all group">
<div class="w-14 h-14 bg-primary/10 rounded-xl flex items-center justify-center text-primary mb-6 group-hover:bg-primary group-hover:text-w
<span class="material-symbols-outlined text-3xl">verified_user</span>
</div>
<h3 class="text-xl font-bold mb-3 text-slate-900 dark:text-slate-100">Verified Listings</h3>
<p class="text-slate-600 dark:text-slate-400">Every job post is vetted for authenticity and transparency to ensure your safety and career g
</div>
<div class="p-8 rounded-2xl bg-white dark:bg-slate-800 border border-primary/10 shadow-sm hover:shadow-xl transition-all group">
<div class="w-14 h-14 bg-primary/10 rounded-xl flex items-center justify-center text-primary mb-6 group-hover:bg-primary group-hover:text-w
<span class="material-symbols-outlined text-3xl">public</span>
</div>
<h3 class="text-xl font-bold mb-3 text-slate-900 dark:text-slate-100">Global Reach</h3>
```



```
<html lang="en"><head>
<body class="bg-background-light dark:bg-background-dark text-slate-900 dark:text-slate-100 font-display">
<div class="relative flex min-h-screen w-full flex-col overflow-x-hidden">
<main class="flex-1">
<section class="py-20 px-6 lg:px-20">
</div>
<h3 class="text-xl font-bold mb-3 text-slate-900 dark:text-slate-100">Global Reach</h3>
<p class="text-slate-600 dark:text-slate-400">Connect with top firms worldwide, from local startups to Fortune 500 giants in every timezone
</div>
<div class="p-8 rounded-2xl bg-white dark:bg-slate-800 border border-primary/10 shadow-sm hover:shadow-xl transition-all group">
<div class="w-14 h-14 bg-primary/10 rounded-xl flex items-center justify-center text-primary mb-6 group-hover:bg-primary group-hover:text-w
<span class="material-symbols-outlined text-3xl">construction</span>
</div>
<h3 class="text-xl font-bold mb-3 text-slate-900 dark:text-slate-100">Career Tools</h3>
<p class="text-slate-600 dark:text-slate-400">Access exclusive resume builders, interview preparation guides, and industry salary insights.
</div>
</div>
</div>
</section>
<!-- Call to Action -->
<section class="px-6 lg:px-20 pb-20">
<div class="max-w-7xl mx-auto rounded-3xl bg-primary px-8 py-16 text-center text-white relative overflow-hidden shadow-2xl shadow-primary/3
<div class="absolute top-0 right-0 p-20 bg-white/10 rounded-full blur-3xl -mr-20 -mt-20"></div>
<div class="absolute bottom-0 left-0 p-20 bg-black/10 rounded-full blur-3xl -ml-20 -mb-20"></div>
<div class="relative z-10 space-y-8">
<h2 class="text-4xl lg:text-5xl font-black">Ready to make your next move?</h2>
<p class="text-white/80 text-lg max-w-xl mx-auto">Join thousands of professionals finding their career path every day through HireSphere.</
<div class="flex flex-col sm:flex-row justify-center gap-4">
<button class="bg-white text-primary font-bold px-10 py-4 rounded-xl hover:bg-slate-50 hover:scale-105 transition-all">Post a Job</button>
<a button class="bg-slate-900 text-white font-bold px-10 py-4 rounded-xl hover:bg-black hover:scale-105 transition-all" href="p23home3.html
</div>
</div>
</div>
</section>
</main>
<!-- Footer -->
<footer class="bg-white dark:bg-slate-900 border-t border-primary/10 pt-16 pb-8 px-6 lg:px-20">
<div class="max-w-7xl mx-auto">
<div class="grid grid-cols-2 md:grid-cols-4 lg:grid-cols-5 gap-12 mb-16">
<div class="col-span-2 space-y-6">
<div class="flex items-center gap-2 text-primary">
<span class="material-symbols-outlined text-3xl font-bold">corporate_fare</span>
<h2 class="text-xl font-extrabold tracking-tight text-slate-900 dark:text-slate-100">HireSphere</h2>
</div>
<p class="text-slate-600 dark:text-slate-400 max-w-xs">Building the bridge between your passion and your profession. The future of hiring s
<div class="flex gap-4">
<a class="w-10 h-10 rounded-full bg-primary/10 flex items-center justify-center text-primary hover:bg-primary hover:text-white transition-a
<span class="material-symbols-outlined text-xl">share</span>
```

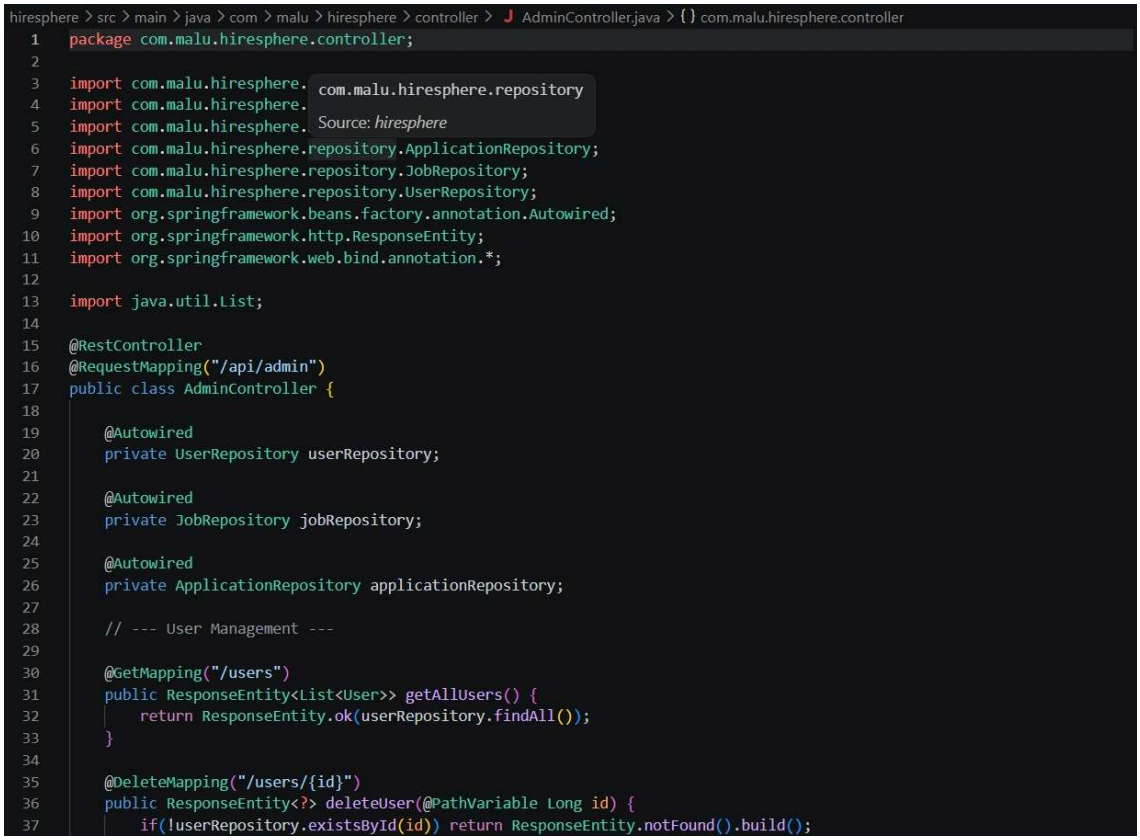


```
<footer class="bg-white dark:bg-slate-900 border-t border-primary/10 pt-16 pb-8 px-6 lg:px-20">
<div class="max-w-7xl mx-auto">
<span class="material-symbols-outlined text-xl">public</span>
</a>
</div>
</div>
<div class="space-y-4">
<h4 class="font-bold text-slate-900 dark:text-slate-100">For Candidates</h4>
<ul class="space-y-2 text-sm text-slate-600 dark:text-slate-400">
<li><a class="hover:text-primary transition-colors" href="#">Browse Jobs</a></li>
<li><a class="hover:text-primary transition-colors" href="#">Career Advice</a></li>
<li><a class="hover:text-primary transition-colors" href="#">Resume Builder</a></li>
</ul>
</div>
<div class="space-y-4">
<h4 class="font-bold text-slate-900 dark:text-slate-100">For Employers</h4>
<ul class="space-y-2 text-sm text-slate-600 dark:text-slate-400">
<li><a class="hover:text-primary transition-colors" href="#">Post a Job</a></li>
<li><a class="hover:text-primary transition-colors" href="#">Recruitment Solutions</a></li>
<li><a class="hover:text-primary transition-colors" href="#">Pricing</a></li>
</ul>
</div>
<div class="space-y-4">
<h4 class="font-bold text-slate-900 dark:text-slate-100">Company</h4>
<ul class="space-y-2 text-sm text-slate-600 dark:text-slate-400">
<li><a class="hover:text-primary transition-colors" href="#">About Us</a></li>
<li><a class="hover:text-primary transition-colors" href="#">Contact</a></li>
<li><a class="hover:text-primary transition-colors" href="#">Privacy Policy</a></li>
</ul>
</div>
</div>
<div class="pt-8 border-t border-slate-100 dark:border-slate-800 text-center text-xs text-slate-500">
<p>© 2026 HireSphere Recruitment Services. All rights reserved.</p>
</div>
</div>
</footer>
</div>
</body></html>
```

Backend

1. HireSphereApplication.java

The HireSphereApplication.java file is the main entry point of the Spring Boot application. It contains the main() method, which is responsible for launching the entire backend system. When this file is executed, Spring Boot initializes the application, starts the embedded server (such as Tomcat), and automatically configures all required components using annotations like @SpringBootApplication. This file enables features like auto-configuration, component scanning, and dependency injection, ensuring that all parts of the project work together seamlessly.



2. AdminController.java

The AdminController.java class is responsible for handling all admin-related operations in the backend. It acts as a REST controller that receives HTTP requests related to administrative tasks such as managing users, approving profiles, monitoring system data, or handling job postings. Using annotations like @RestController and @RequestMapping, it defines API endpoints that allow the admin to interact with the system. This controller connects the frontend admin panel with the backend logic and ensures that admin actions are processed securely and

efficiently.

```java
package com.malu.hiresphere.controller;

import com.malu.hiresphere.model.Application;
import com.malu.hiresphere.model.Job;
import com.malu.hiresphere.model.User;
import com.malu.hiresphere.repository.UserRepository;
import com.malu.hiresphere.service.JobService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;

import java.util.List;
import java.util.Map;

@RestController
@RequestMapping("/api/company")
@CrossOrigin(origins = "*")
public class CompanyController {

    @Autowired
    private JobService jobService;

    @Autowired
    private UserRepository userRepository;

    @Autowired
    private com.malu.hiresphere.repository.ApplicationRepository applicationRepository;

    @Autowired
    private com.malu.hiresphere.repository.CandidateProfileRepository profileRepository;

    @PostMapping("/{companyId}/jobs")
    public ResponseEntity<?> postJob(@PathVariable Long companyId, @RequestBody Job job) {
        User company = userRepository.findById(companyId).orElseThrow();
        job.setCompany(company);
        Job savedJob = jobService.createJob(job);
        return ResponseEntity.ok(savedJob);
```

3. CompanyController.java

The CompanyController.java class handles all company-related operations in the backend system. It is a REST controller that processes HTTP requests from companies, such as creating job postings, updating company profiles, and viewing applicants. Using annotations like @RestController and @RequestMapping, it defines API endpoints that allow companies to interact with the platform. This controller acts as a bridge between the frontend company interface and the backend services, ensuring that company data and actions are managed efficiently.

```java
package com.malu.hiresphere.controller;

import com.malu.hiresphere.model.Application;
import com.malu.hiresphere.model.Job;
import com.malu.hiresphere.model.User;
import com.malu.hiresphere.repository.UserRepository;
import com.malu.hiresphere.service.JobService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;

import java.util.List;
import java.util.Map;

@RestController
@RequestMapping("/api/company")
@CrossOrigin(origins = "*")
public class CompanyController {

    @Autowired
    private JobService jobService;

    @Autowired
    private UserRepository userRepository;

    @Autowired
    private com.malu.hiresphere.repository.ApplicationRepository applicationRepository;

    @Autowired
    private com.malu.hiresphere.repository.CandidateProfileRepository profileRepository;

    @PostMapping("/{companyId}/jobs")
    public ResponseEntity<?> postJob(@PathVariable Long companyId, @RequestBody Job job) {
        User company = userRepository.findById(companyId).orElseThrow();
        job.setCompany(company);
        Job savedJob = jobService.createJob(job);
        return ResponseEntity.ok(savedJob);
```

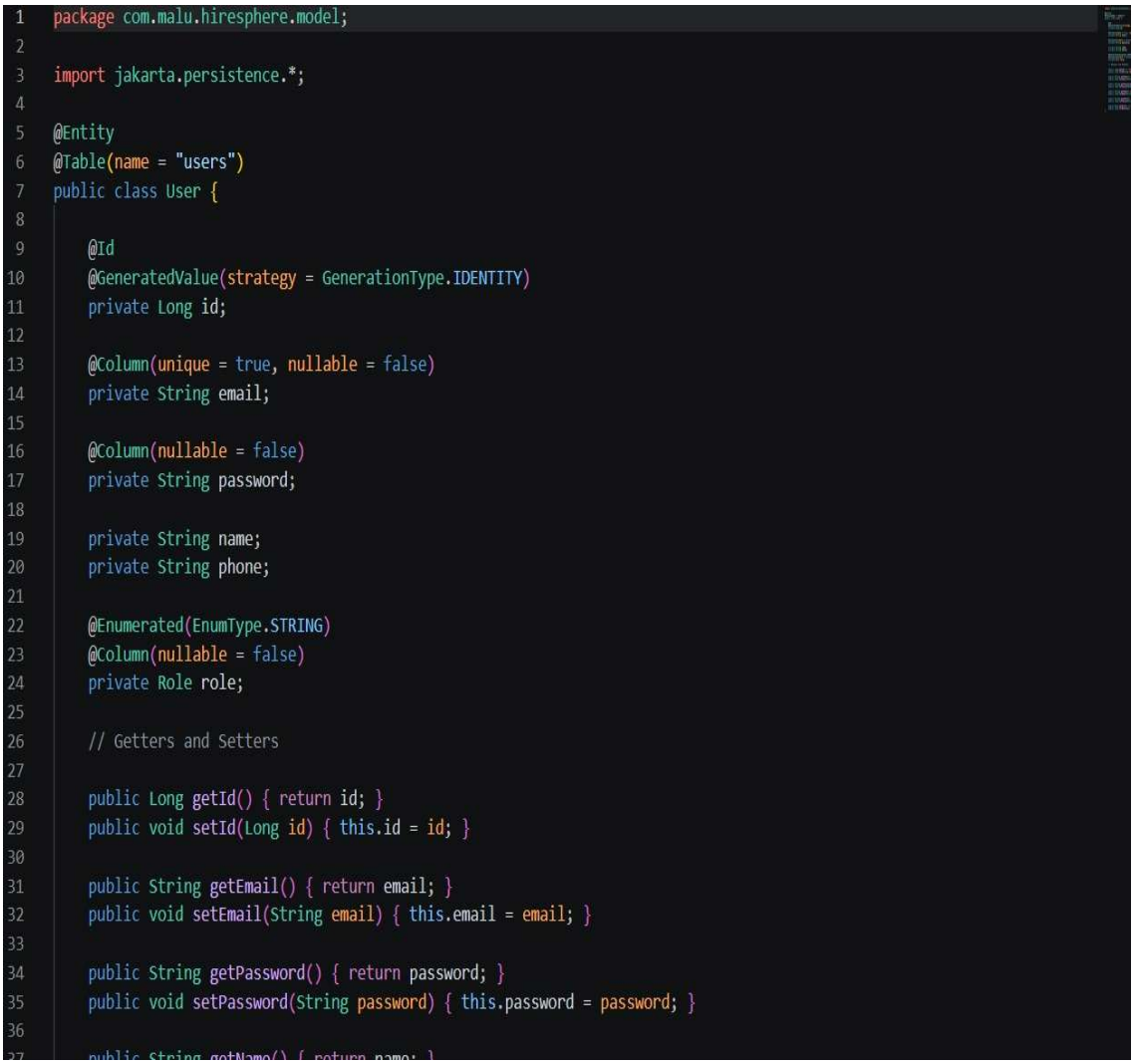
4. ResumeAnalyzerService.java

The ResumeAnalyzerService.java class is responsible for processing and analyzing uploaded resumes in the system. It contains the core business logic to extract important information such as skills, qualifications, and experience from resumes. This service may use techniques like text parsing, keyword extraction, or basic machine learning to evaluate candidate profiles. It helps in comparing candidate skills with job requirements and supports features like skill matching, ranking, and recommendations. By handling all analysis logic in one place, it keeps the application organized and improves scalability.



5 .CandidateProfile.java

The CandidateProfile.java class represents the detailed profile of a job seeker in the system. It is typically defined as an entity that maps to a database table, storing information such as personal details, education, skills, experience, and resume data. This class acts as a structured model for managing candidate information and is used throughout the application for operations like profile creation, updates, and job matching. It helps organize user data efficiently and plays a key role in connecting candidates with suitable job opportunities.



```java
package com.malu.hiresphere.model;

import jakarta.persistence.*;

@Entity
@Table(name = "users")
public class User {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(unique = true, nullable = false)
    private String email;

    @Column(nullable = false)
    private String password;

    private String name;
    private String phone;

    @Enumerated(EnumType.STRING)
    @Column(nullable = false)
    private Role role;

    // Getters and Setters

    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }

    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }

    public String getPassword() { return password; }
    public void setPassword(String password) { this.password = password; }

    public String getName() { return name; }
```

CHAPTER 6

OUTPUT

The proposed solution was successfully implemented and tested, demonstrating an effective system for assessing and matching candidates based on their skills. The system accurately analyzed resumes, conducted skills assessment tests, and produced compatibility scores to match job seekers to appropriate job openings. By automating many of the manual processes traditionally involved in recruiting, the recruitment process has become more efficient and has reduced the number of mismatches in the hiring process. Recruiters can now use the ranking feature provided by the application to quickly identify the best-fit job candidates. Furthermore, the application identifies skill gaps and provides recommendations to assist job seekers with developing their skills. The anonymity of the applicant's screening process provides equality for all candidates; in summary, the system has successfully fulfilled its objectives as an efficient, accurate, and unbiased method for recruiting candidates to fill job openings

4.1 User home page

This screen shows the home page of the employment website, which is meant to provide a clean, easy-to-use experience. There is a central search area for users to enter job title(s), keyword(s), or location where they would like to find job opportunities. The page has been designed to help users find jobs that match their skills and interests. Additionally, there is a section for top hiring companies, giving users a simple way to connect with well-known companies and learn about job openings. The navigation menu allows users to easily navigate between the site sections (e.g., Home, Companies, About Us, Sign Up, Log In, and Profile) for an uninterrupted user experience. Overall, the home page seamless blends functionality with simplicity to improve the user's overall experience.

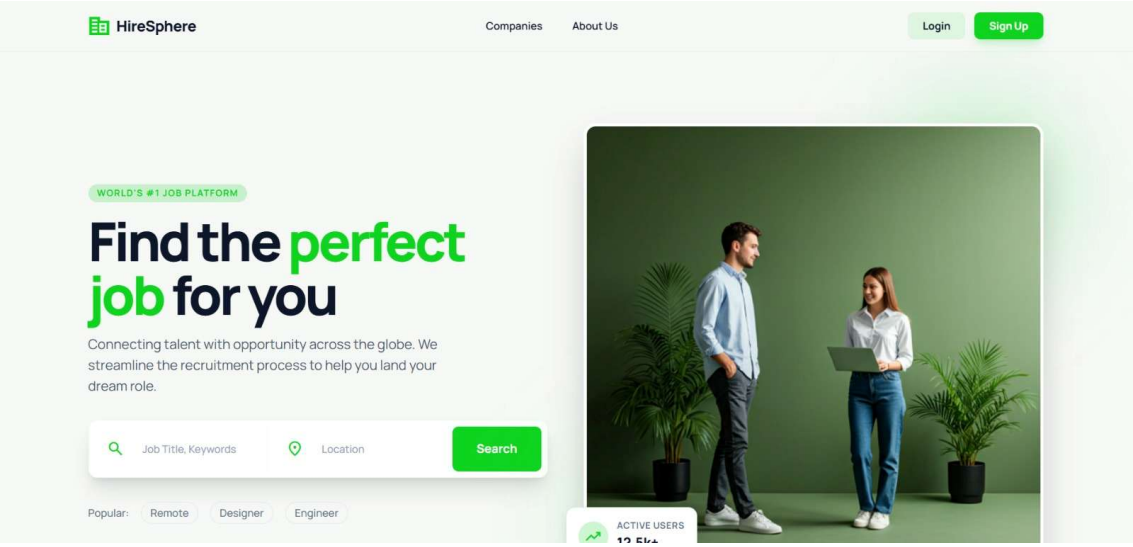


Fig 4.1 : User home page

4.2 Signup page

Above, you see what is called a “user registration page.” This is where new users create their account to use the features of the system. In addition to creating an account, the user will complete a simple form that collects their first name, last name, email address, phone number, and password to verify their identity as well as authenticate them less securely than other systems. The user registration page has a very simple and clean structure that allows users to complete their information correctly without confusion. This module allows users to complete the registration process quickly, which is helpful for them accessing other parts of the platform such as job searching, skill assessments, and personalized recommendations. Overall this is a very important module for the user's initial experience with the platform.

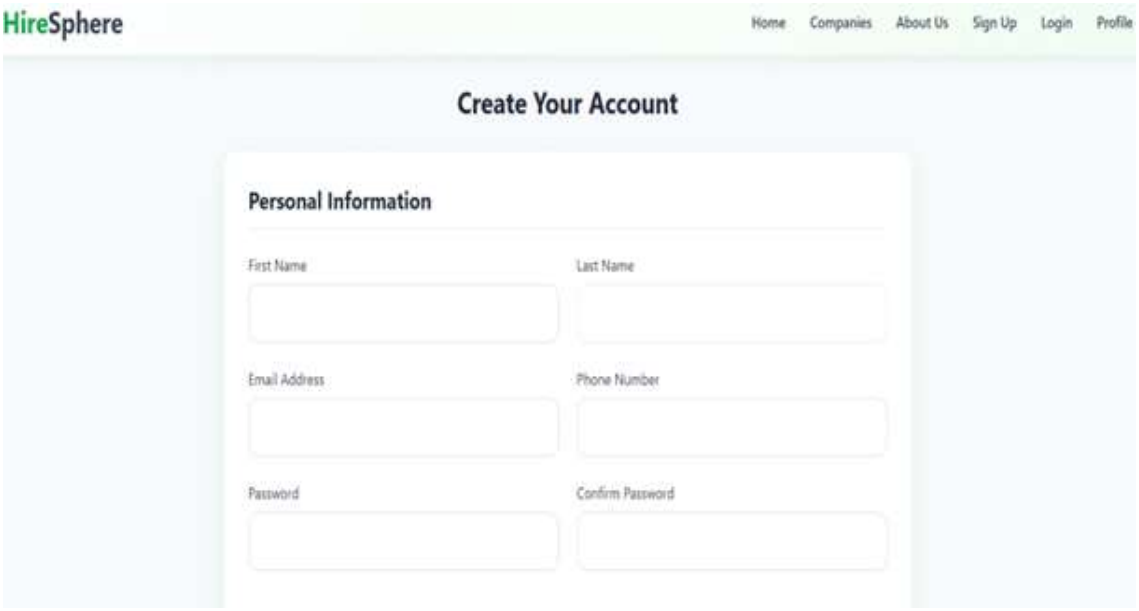


Fig 4.2 : Signup page

4.3 Candidate Registration Portal

Users can complete their profiles for accessing the services offered on the Candidate Registration Portal, represented in the form of an above screen shot of the Candidate Registration Portal. As multiple sections of the page are available to users, such as Personal Profile, Academic Credentials, etc., see below for examples of data being collected by section; candidates will be required to enter detailed data, including name, email, telephone number and password. Detailed data can be entered and managed simply from one to another using a step-by-step approach due to the organized structure of the page. Ultimately, this module is very important when collecting comprehensive information about candidates in order to conduct skills assessments and to match candidates to job positions with personalized recommendations for a greater degree of accuracy in the recruitment process.

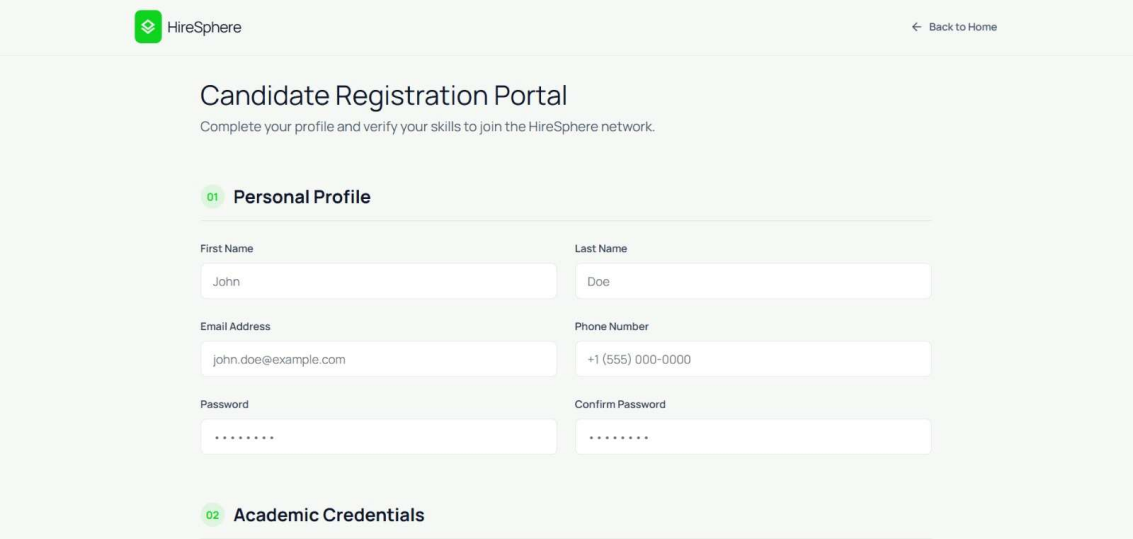


Fig 4.3 : Candidate Registration Portal

4.4 Profile page

This screen is the user profile area of a skill-based job placement system (job placement systems that focus on the skills that people have rather than only the high-level status of employers), where potential employees and employers can access their personal information and professional history. The profile shows the following (for example, your profile): the person's first and last names, e-mail address, phone number and geographical location (city/state); edit profile; further information about the potential employee such as education history (college, etc.); certificates or diplomas earned (i.e.: a high school diploma, etc.) and can (and should) be made visible to an employer; the ability to mark your privacy status about your profile by concealing (or masking) identifying characteristics like your birthdate; and there are organized sections/rows so that it is easy to find all of this information together without having to search for it; and this makes it possible to manage your profile effectively for the purpose of sending correct information regarding the placement of employees in positions by means of being able to provide accurate data to support position decisions made by hiring authorities.

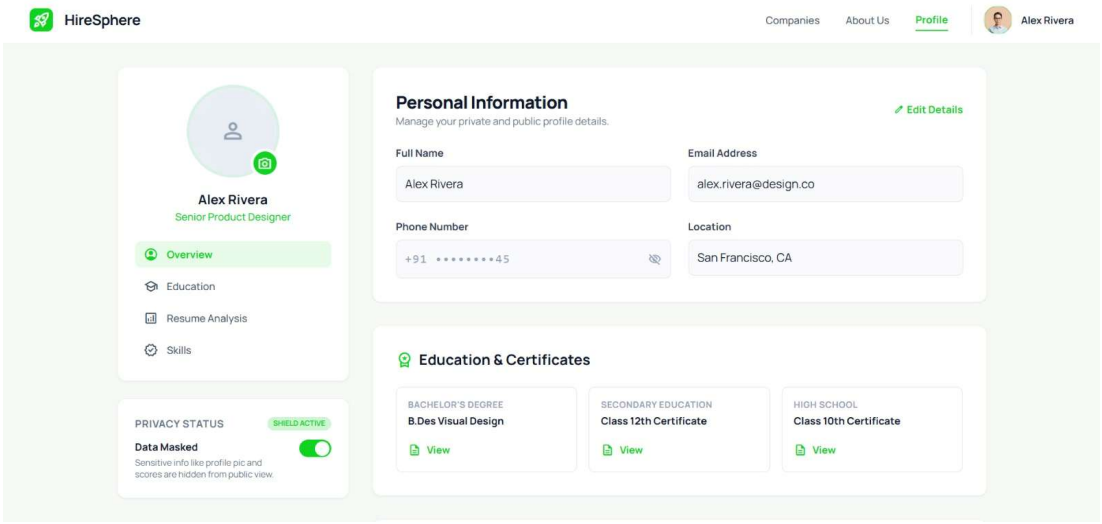


Fig 4.4 : Profile page

4.5 Resume Analysis and Skills Assessment page

This screen illustrates the AI Resume Analysis and Skills Assessment portion of our application, which enables the user to upload their own resume(s) and receive an evaluation based on numerous criteria/standards. Once the resume has been evaluated, the system provides two outputs - a quality score (indicating how closely the uploaded resume conforms to industry standards) and recommendations for improvement (such as keyword optimization). The system also provides the user with a list of verified skills along with assessment results, as well as information regarding their level of performance by skill area (i.e., lower performing areas need improvement). As well, the user has the option to continue taking assessments that will allow them to improve their quality rating and enhance their overall profile. In summary, this module provides an essential service by validating candidates' skills and improving the accuracy of job matching through the use of data-driven results**.**

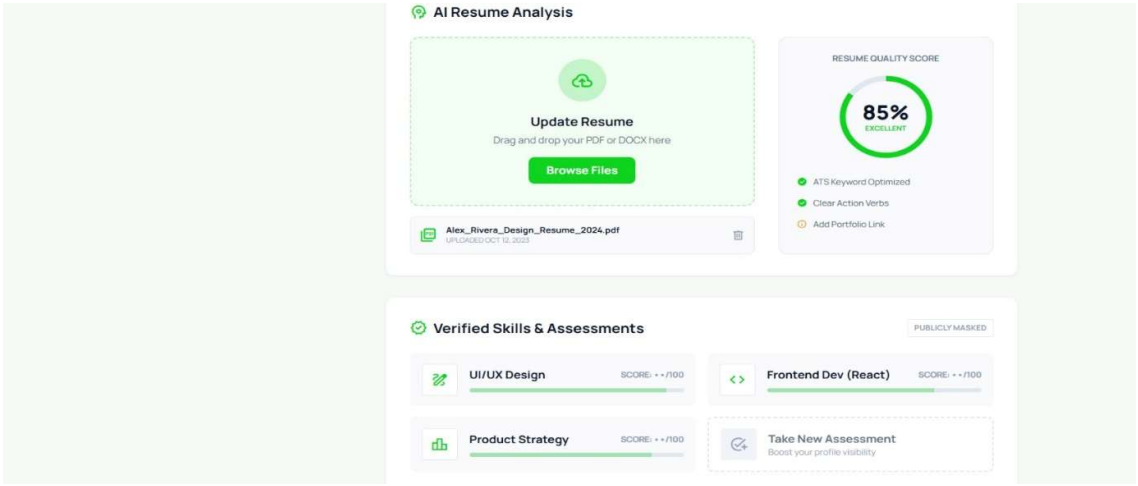


Fig 4.5 : Resume Analysis and Skills Assessment page

4.6 Companies Directory Page

The HireSphere platform's Companies Directory Page provides users with an easy way to search for various Companies based on type (e.g., Software, Service, and Consulting). Users have a search box so that they can quickly locate a given Company and can view each relevant Company with its name, type of Industry, and a 'View Jobs' button, which they can use to browse through each Company's available Job openings. Combined with the organization of Companies, this Page makes it easy to search for Job openings.

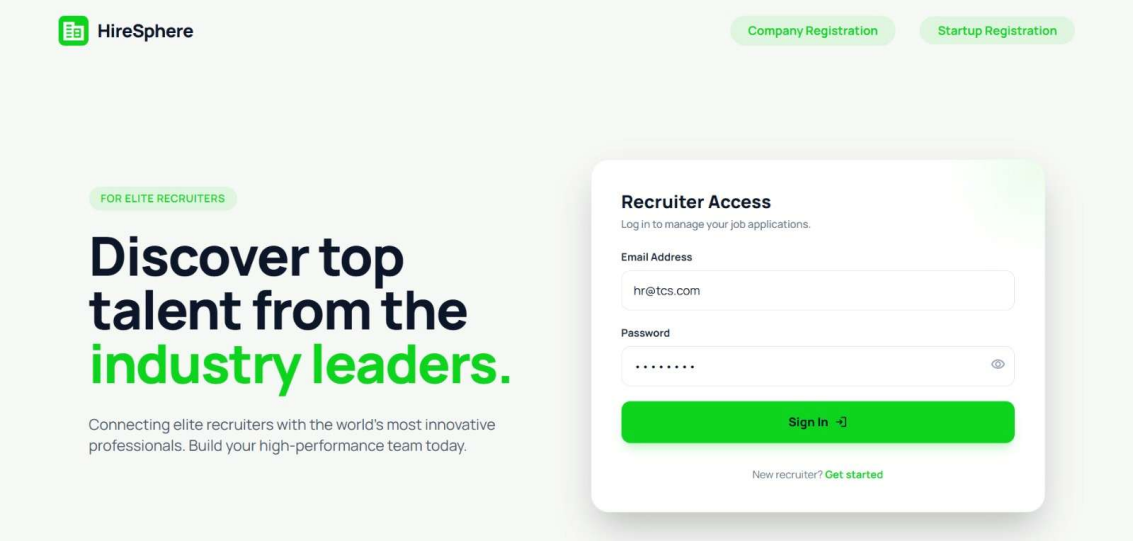


Fig 4.6 : Companies Directory Page

CHAPTER 7

CONCLUSION AND FUTURE WORKS

The skill-based job placement system that was proposed addresses many of the shortcomings of traditional recruitment methods, by providing an innovative, streamlined and non-discriminatory means of employment selection. This new way of selecting employees places emphasis solely on measuring the applicant's abilities and performance as opposed to solely relying on an individual's résumé. This shift from reliance on the previous way of recruiting has yielded a more definitive and reliable method of employing individuals. The incorporation of additional elements, such as résumé analysis, skills assessments, job matching and skill gap identification, into the comprehensive job placement system creates a complete resource for both employers and top talent. Job seekers can determine areas in which they excel, as well as those in which they can improve their skills and qualities, and be matched with an appropriate position based on the information contained in their profile and additional criteria defined by the employer. Employers can quickly determine which applicants are most likely to qualify for a specific job due to their compatibility score and rank, making it easy for employers to identify and select candidates who have the highest probability of being hired. The implementation of anonymous screening in the initial stages of the recruitment process further enhances the integrity of the entire process due to the minimization of bias during these early stages of the hiring process. As a result, this system accomplishes its desired goals of being a completely transparent, accurate and effective method for recruiting and hiring candidates.

In order to enhance the system in the future, advanced technologies such as artificial intelligence and machine learning will be incorporated into it to provide accurate assessments of skills and job matches. The introduction of real-time interview modules (video-based interviews and AI-enabled evaluation) will provide further accuracy for assessing candidates. Furthermore, the platform can also support a wide variety of languages for the accessibility of a larger audience. Integrating with previous job portals and professional networking sites increases job availability and reach. Advanced data analytics and visualization tools can be used to provide recruiters with improved insights. The development of a mobile application will improve accessibility and engagement for users while enhanced security mechanisms (e.g., multi-factor authentication) can protect user data. With ongoing enhancements and upgrades, the system will continue to grow to be an all-inclusive and intelligent recruitment solution for the modern workforce.

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